

The map δ is an isomorphism when restricted to the copy of $H^n(Y; R)$ corresponding to $\{0\} \times Y$. This copy of $H^n(Y; R)$ consists of elements of the form $1_0 \times \beta$ where $1_0 \in H^0(\partial I; R)$ is represented by the cocycle that is 1 on $0 \in \partial I$ and 0 on $1 \in \partial I$. By the commutative square above, $\delta(1_0 \times \beta) = \delta(1_0) \times \beta$. The element $\delta(1_0)$ is a generator of $H^1(I, \partial I; R)$, by the case that Y is a point. Any other generator α is a scalar multiple of $\delta(1_0)$ by a unit of R , so this shows the map $\beta \mapsto \alpha \times \beta$ is an isomorphism.

An equivalent statement is that the map $H^n(Y; R) \rightarrow H^{n+1}(S^1 \times Y, \{s_0\} \times Y; R)$, $\beta \mapsto \alpha \times \beta$, is an isomorphism, with α now a generator of $H^1(S^1, s_0; R)$. Via the long exact sequence of the pair $(S^1 \times Y, \{s_0\} \times Y)$, this implies that the map

$$H^{n+1}(Y; R) \times H^n(Y; R) \rightarrow H^{n+1}(S^1 \times Y; R), \quad (\beta_1, \beta_2) \mapsto 1 \times \beta_1 + \alpha \times \beta_2$$

is an isomorphism, with α a generator of $H^1(S^1; R)$. Specializing to the case of the n -torus, we conclude by induction on n that $H^k(T^n; R)$ has the structure described at the beginning of the example.

We can use this calculation to deduce a fact that will be used shortly in the calculation of cup products in projective spaces. Writing $n = i + j$, the cube I^n is the product $I^i \times I^j$, and the assertion is that the cross product of generators of $H^i(I^i, \partial I^i; R)$ and $H^j(I^j, \partial I^j; R)$ is a generator of $H^n(I^n, \partial I^n; R)$, where we are using the first of the following three cross products:

$$\begin{aligned} H^i(I^i, \partial I^i; R) \times H^j(I^j, \partial I^j; R) &\xrightarrow{\times} H^n(I^n, \partial I^n; R) \\ H^i(T^i, \dot{T}^i; R) \times H^j(T^j, \dot{T}^j; R) &\xrightarrow{\times} H^n(T^n, \dot{T}^n; R) \\ H^i(T^i; R) \times H^j(T^j; R) &\xrightarrow{\times} H^n(T^n; R) \end{aligned}$$

In the second cross product, the dots denote deletion of the top-dimensional cell. All three cross products are equivalent. This is evident for the first two, thinking of the torus as a quotient of a cube. For the second two, note that all cellular boundary maps for T^n with $\mathbb{Z}$ coefficients must be trivial, otherwise the cohomology groups would be smaller than computed above. Hence all cellular coboundary maps with arbitrary coefficients are zero, and the map $H^n(T^n, \dot{T}^n; R) \rightarrow H^n(T^n; R)$ is an isomorphism. The corresponding results for T^i and T^j are of course true as well.

Since cross product is associative, the earlier calculation shows that for the last of the three cross products above, the cross product of generators is a generator, so this is also true for the first cross product.

The Cohomology Ring

Since cup product is associative and distributive, it is natural to try to make it the multiplication in a ring structure on the cohomology groups of a space X . This is easy to do if we simply define $H^*(X; R)$ to be the direct sum of the groups $H^n(X; R)$. Elements of $H^*(X; R)$ are finite sums $\sum_i \alpha_i$ with $\alpha_i \in H^i(X; R)$, and the product of

two such sums is defined to be $(\sum_i \alpha_i)(\sum_j \beta_j) = \sum_{i,j} \alpha_i \beta_j$. It is routine to check that this makes $H^*(X; R)$ into a ring, with identity if R has an identity. Similarly, $H^*(X, A; R)$ is a ring via the relative cup product. Taking scalar multiplication by elements of R into account, these rings can also be regarded as R -algebras.

For example, the calculations in Example 3.8 or 3.9 above show that $H^*(\mathbb{RP}^2; \mathbb{Z}_2)$ consists of the polynomials $a_0 + a_1 \alpha + a_2 \alpha^2$ with coefficients $a_i \in \mathbb{Z}_2$, so $H^*(\mathbb{RP}^2; \mathbb{Z}_2)$ is the quotient $\mathbb{Z}_2[\alpha]/(\alpha^3)$ of the polynomial ring $\mathbb{Z}_2[\alpha]$ by the ideal generated by α^3 .

This example illustrates how $H^*(X; R)$ often has a more compact description than the sequence of individual groups $H^n(X; R)$, so there is a certain economy in the change of scale that comes from regarding all the groups $H^n(X; R)$ as part of a single object $H^*(X; R)$.

Adding cohomology classes of different dimensions to form $H^*(X; R)$ is a convenient formal device, but it has little topological significance. One always regards the cohomology ring as a **graded ring**: a ring A with a decomposition as a sum $\bigoplus_{k \geq 0} A_k$ of additive subgroups A_k such that the multiplication takes $A_k \times A_\ell$ to $A_{k+\ell}$. To indicate that an element $a \in A$ lies in A_k we write $|a| = k$. This applies in particular to elements of $H^k(X; R)$. Some authors call $|a|$ the ‘degree’ of a , but we will use the term ‘dimension’ which is more geometric and avoids potential confusion with the degree of a polynomial.

Among the simplest graded rings are polynomial rings $R[\alpha]$ and their truncated versions $R[\alpha]/(\alpha^n)$, consisting of polynomials of degree less than n . The example we have seen is $H^*(\mathbb{RP}^2; \mathbb{Z}_2) \approx \mathbb{Z}_2[\alpha]/(\alpha^3)$. Generalizing this, we have:

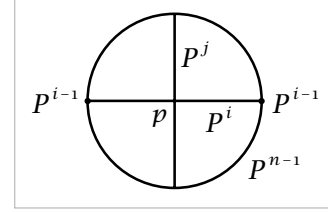
Theorem 3.12. $H^*(\mathbb{RP}^n; \mathbb{Z}_2) \approx \mathbb{Z}_2[\alpha]/(\alpha^{n+1})$ and $H^*(\mathbb{RP}^\infty; \mathbb{Z}_2) \approx \mathbb{Z}_2[\alpha]$, where $|\alpha| = 1$. In the complex case, $H^*(\mathbb{CP}^n; \mathbb{Z}) \approx \mathbb{Z}[\alpha]/(\alpha^{n+1})$ and $H^*(\mathbb{CP}^\infty; \mathbb{Z}) \approx \mathbb{Z}[\alpha]$ where $|\alpha| = 2$.

This turns out to be a quite important result, and it can be proved in a number of different ways. The proof we give here consists of a direct reduction to the relative cup product calculation in Example 3.11 above. Another proof using Poincaré duality will be given in Example 3.40. A third proof is contained in §4.D as an application of the Gysin sequence.

Proof: Let us do the case of $\mathbb{RP}^n$ first. To simplify notation we abbreviate $\mathbb{RP}^n$ to P^n and we let the coefficient group $\mathbb{Z}_2$ be implicit. Since the inclusion $P^{n-1} \hookrightarrow P^n$ induces an isomorphism on H^i for $i \leq n-1$, it suffices by induction on n to show that the cup product of a generator of $H^{n-1}(P^n)$ with a generator of $H^1(P^n)$ is a generator of $H^n(P^n)$. It will be no more work to show more generally that the cup product of a generator of $H^i(P^n)$ with a generator of $H^{n-i}(P^n)$ is a generator of $H^n(P^n)$. As a further notational aid, we let $j = n - i$, so $i + j = n$.

The proof uses some of the geometric structure of P^n . Recall that P^n consists of nonzero vectors $(x_0, \dots, x_n) \in \mathbb{R}^{n+1}$ modulo multiplication by nonzero scalars. In-

side P^n is a copy of P^i represented by vectors whose last j coordinates $x_{i+1}, \dots, x_n$ are zero. We also have a copy of P^j represented by points whose first i coordinates $x_0, \dots, x_{i-1}$ are zero. The intersection $P^i \cap P^j$ is a single point p , represented by vectors whose only nonzero coordinate is x_i . Let U be the subspace of P^n represented by vectors with nonzero coordinate x_i . Each point in U may be represented by a unique vector with $x_i = 1$ and the other n coordinates arbitrary, so U is homeomorphic to $\mathbb{R}^n$, with p corresponding to 0 under this homeomorphism.



We can write this $\mathbb{R}^n$ as $\mathbb{R}^i \times \mathbb{R}^j$, with $\mathbb{R}^i$ as the coordinates $x_0, \dots, x_{i-1}$ and $\mathbb{R}^j$ as the coordinates $x_{i+1}, \dots, x_n$. In the figure P^n is represented as a disk with antipodal points of its boundary sphere identified to form a $P^{n-1} \subset P^n$ with $U = P^n - P^{n-1}$ the interior of the disk.

Consider the diagram

$$\begin{array}{ccccc}
 H^i(P^n) \times H^j(P^n) & \xrightarrow{\quad \smile \quad} & H^n(P^n) \\
 \uparrow & & \uparrow \\
 H^i(P^n, P^n - P^j) \times H^j(P^n, P^n - P^i) & \xrightarrow{\quad \smile \quad} & H^n(P^n, P^n - \{p\}) \\
 \downarrow & & \downarrow \\
 H^i(\mathbb{R}^n, \mathbb{R}^n - \mathbb{R}^j) \times H^j(\mathbb{R}^n, \mathbb{R}^n - \mathbb{R}^i) & \xrightarrow{\quad \smile \quad} & H^n(\mathbb{R}^n, \mathbb{R}^n - \{0\})
 \end{array}$$

which commutes by naturality of cup product. The lower cup product map takes generator cross generator to generator, as we showed in Example 3.11 above in the equivalent situation of a product of cubes. The same will be true for the top row if the four vertical maps are isomorphisms, so this is what remains to be proved.

The lower map in the right column is an isomorphism by excision. For the upper map in this column, the fact that $P^n - \{p\}$ deformation retracts to a P^{n-1} gives an isomorphism $H^n(P^n, P^n - \{0\}) \approx H^n(P^n, P^{n-1})$ via the five-lemma applied to the long exact sequences for these pairs. And $H^n(P^n, P^{n-1}) \approx H^n(P^n)$ by cellular cohomology.

To see that the vertical maps in the left column are isomorphisms we use the following commutative diagram:

$$\begin{array}{ccccccc}
 H^i(P^n) & \longleftarrow & H^i(P^n, P^{i-1}) & \longleftarrow & H^i(P^n, P^n - P^j) & \longrightarrow & H^i(\mathbb{R}^n, \mathbb{R}^n - \mathbb{R}^j) \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 H^i(P^i) & \longleftarrow & H^i(P^i, P^{i-1}) & \longleftarrow & H^i(P^i, P^i - \{p\}) & \longrightarrow & H^i(\mathbb{R}^i, \mathbb{R}^i - \{0\})
 \end{array}$$

If we can show all these maps are isomorphisms, then the same argument will apply with i and j interchanged, and the proof for $\mathbb{R}P^n$ will be finished.

The left-hand square consists of isomorphisms by cellular cohomology. The right-hand vertical map is obviously an isomorphism. The lower right horizontal map is an isomorphism by excision, and the map to the left of this is an isomorphism since $P^i - \{p\}$ deformation retracts onto P^{i-1} . The remaining maps will be isomorphisms if the middle map in the upper row is an isomorphism. And this map is in fact

an isomorphism because $P^n - P^j$ deformation retracts onto P^{i-1} by the following argument. The subspace $P^n - P^j \subset P^n$ consists of points represented by vectors $v = (x_0, \dots, x_n)$ with at least one of the coordinates $x_0, \dots, x_{i-1}$ nonzero. The formula $f_t(v) = (x_0, \dots, x_{i-1}, tx_i, \dots, tx_n)$ for t decreasing from 1 to 0 gives a well-defined deformation retraction of $P^n - P^j$ onto P^{i-1} since $f_t(\lambda v) = \lambda f_t(v)$ for scalars $\lambda \in \mathbb{R}$.

The case of $\mathbb{R}P^\infty$ follows from the finite-dimensional case since the inclusion $\mathbb{R}P^n \hookrightarrow \mathbb{R}P^\infty$ induces isomorphisms on $H^i(-; \mathbb{Z}_2)$ for $i \leq n$ by cellular cohomology.

Complex projective spaces are handled in precisely the same way, using $\mathbb{Z}$ coefficients and replacing each H^k by H^{2k} and $\mathbb{R}$ by $\mathbb{C}$. $\square$

There are also quaternionic projective spaces $\mathbb{H}P^n$ and $\mathbb{H}P^\infty$, defined exactly as in the complex case, with CW structures of the form $e^0 \cup e^4 \cup e^8 \cup \dots$. Associativity of quaternion multiplication is needed for the identification $v \sim \lambda v$ to be an equivalence relation, so the definition does not extend to octonionic projective spaces, though there is an octonionic projective plane $\mathbb{O}P^2$ that will be defined in §4.3. The cup product structure in quaternionic projective spaces is just like that in complex projective spaces, except that the generator is 4-dimensional:

$$H^*(\mathbb{H}P^\infty; \mathbb{Z}) \approx \mathbb{Z}[\alpha] \quad \text{and} \quad H^*(\mathbb{H}P^n; \mathbb{Z}) \approx \mathbb{Z}[\alpha]/(\alpha^{n+1}), \quad \text{with } |\alpha| = 4$$

The same proof as in the real and complex cases works as well in this case.

The cup product structure for $\mathbb{R}P^\infty$ with $\mathbb{Z}$ coefficients can easily be deduced from the cup product structure with $\mathbb{Z}_2$ coefficients, as follows. In general, a ring homomorphism $R \rightarrow S$ induces a ring homomorphism $H^*(X, A; R) \rightarrow H^*(X, A; S)$. In the case of the projection $\mathbb{Z} \rightarrow \mathbb{Z}_2$ we get for $\mathbb{R}P^\infty$ an induced chain map of cellular cochain complexes with $\mathbb{Z}$ and $\mathbb{Z}_2$ coefficients:

$$\begin{array}{ccccccccccc} \cdots & \longleftarrow & \mathbb{Z} & \xleftarrow{2} & \mathbb{Z} & \xleftarrow{0} & \mathbb{Z} & \xleftarrow{2} & \mathbb{Z} & \xleftarrow{0} & \mathbb{Z} & \longleftarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\ \cdots & \longleftarrow & \mathbb{Z}_2 & \xleftarrow{0} & \mathbb{Z}_2 & \xleftarrow{0} & \mathbb{Z}_2 & \xleftarrow{0} & \mathbb{Z}_2 & \xleftarrow{0} & \mathbb{Z}_2 & \longleftarrow & 0 \end{array}$$

From this we see that the ring homomorphism $H^*(\mathbb{R}P^\infty; \mathbb{Z}) \rightarrow H^*(\mathbb{R}P^\infty; \mathbb{Z}_2)$ is injective in positive dimensions, with image the even-dimensional part of $H^*(\mathbb{R}P^\infty; \mathbb{Z}_2)$. Alternatively, this could be deduced from the universal coefficient theorem. Hence we have $H^*(\mathbb{R}P^\infty; \mathbb{Z}) \approx \mathbb{Z}[\alpha]/(2\alpha)$ with $|\alpha| = 2$.

The cup product structure in $H^*(\mathbb{R}P^n; \mathbb{Z})$ can be computed in a similar fashion, though the description is a little cumbersome:

$$\begin{aligned} H^*(\mathbb{R}P^{2k}; \mathbb{Z}) &\approx \mathbb{Z}[\alpha]/(2\alpha, \alpha^{k+1}), \quad |\alpha| = 2 \\ H^*(\mathbb{R}P^{2k+1}; \mathbb{Z}) &\approx \mathbb{Z}[\alpha, \beta]/(2\alpha, \alpha^{k+1}, \beta^2, \alpha\beta), \quad |\alpha| = 2, \quad |\beta| = 2k+1 \end{aligned}$$

Here β is a generator of $H^{2k+1}(\mathbb{R}P^{2k+1}; \mathbb{Z}) \approx \mathbb{Z}$. From this calculation we see that the rings $H^*(\mathbb{R}P^{2k+1}; \mathbb{Z})$ and $H^*(\mathbb{R}P^{2k} \vee S^{2k+1}; \mathbb{Z})$ are isomorphic, though with $\mathbb{Z}_2$

coefficients this is no longer true, as the generator $\alpha \in H^1(\mathbb{RP}^{2k+1}; \mathbb{Z}_2)$ has $\alpha^{2k+1} \neq 0$, while $\alpha^{2k+1} = 0$ for the generator $\alpha \in H^1(\mathbb{RP}^{2k} \vee S^{2k+1}; \mathbb{Z}_2)$.

Induced homomorphisms are ring homomorphisms by Proposition 3.10. Here is an example illustrating this fact.

Example 3.13. The isomorphism $H^*(\coprod_{\alpha} X_{\alpha}; R) \xrightarrow{\cong} \prod_{\alpha} H^*(X_{\alpha}; R)$ whose coordinates are induced by the inclusions $i_{\alpha}: X_{\alpha} \hookrightarrow \coprod_{\alpha} X_{\alpha}$ is a ring isomorphism with respect to the usual coordinatewise multiplication in a product ring, because each coordinate function i_{α}^* is a ring homomorphism. Similarly for a wedge sum the isomorphism $\tilde{H}^*(\bigvee_{\alpha} X_{\alpha}; R) \approx \prod_{\alpha} \tilde{H}^*(X_{\alpha}; R)$ is a ring isomorphism. Here we take reduced cohomology to be cohomology relative to a basepoint, and we use relative cup products. We should assume the basepoints $x_{\alpha} \in X_{\alpha}$ are deformation retracts of neighborhoods, to be sure that the claimed isomorphism does indeed hold.

This product ring structure for wedge sums can sometimes be used to rule out splittings of a space as a wedge sum up to homotopy equivalence. For example, consider $\mathbb{CP}^2$, which is S^2 with a cell e^4 attached by a certain map $f: S^3 \rightarrow S^2$. Using homology or just the additive structure of cohomology it is impossible to conclude that $\mathbb{CP}^2$ is not homotopy equivalent to $S^2 \vee S^4$, and hence that f is not homotopic to a constant map. However, with cup products we can distinguish these two spaces since the square of each element of $H^2(S^2 \vee S^4; \mathbb{Z})$ is zero in view of the ring isomorphism $\tilde{H}^*(S^2 \vee S^4; \mathbb{Z}) \approx \tilde{H}^*(S^2; \mathbb{Z}) \oplus \tilde{H}^*(S^4; \mathbb{Z})$, but the square of a generator of $H^2(\mathbb{CP}^2; \mathbb{Z})$ is nonzero by Theorem 3.12.

More generally, cup products can be used to distinguish infinitely many different homotopy classes of maps $S^{4n-1} \rightarrow S^{2n}$ for all $n \geq 1$. This is systematized in the notion of the *Hopf invariant*, which is studied in §4.B.

The natural question of whether the cohomology ring is commutative is answered by the following:

Theorem 3.14. *The identity $\alpha \smile \beta = (-1)^{k\ell} \beta \smile \alpha$ holds for all $\alpha \in H^k(X, A; R)$ and $\beta \in H^{\ell}(X, A; R)$, when R is commutative.*

Taking $\alpha = \beta$, this implies in particular that if α is an element of $H^k(X, A; R)$ with k odd, then $2\alpha^2 = 0$ in $H^{2k}(X, A; R)$. Hence if $H^{2k}(X, A; R)$ has no elements of order two, then $\alpha^2 = 0$. For example, if X is the 2-complex obtained by attaching a disk to S^1 by a map of degree m as in Example 3.9 above, then we can deduce that the square of a generator of $H^1(X; \mathbb{Z}_m)$ is zero if m is odd, and is either zero or the unique element of $H^2(X; \mathbb{Z}_m) \approx \mathbb{Z}_m$ of order two if m is even. As we showed, the square is in fact nonzero when m is even.

A graded ring satisfying the commutativity property of the theorem is usually called simply **commutative** in the context of algebraic topology, in spite of the potential for misunderstanding. In the older literature one finds less ambiguous terms such as *graded commutative*, *anticommutative*, or *skew commutative*.

Proof: Consider first the case $A = \emptyset$. For cochains $\varphi \in C^k(X; R)$ and $\psi \in C^\ell(X, R)$ one can see from the definition that the cup products $\varphi \smile \psi$ and $\psi \smile \varphi$ differ only by a permutation of the vertices of $\Delta^{k+\ell}$. The idea of the proof is to study a particularly nice permutation of vertices, namely the one that totally reverses their order, replacing $[v_0, \dots, v_n]$ by $[v_n, \dots, v_0]$. This has the convenient feature of also reversing the ordering of vertices in any face.

For a singular n -simplex $\sigma: [v_0, \dots, v_n] \rightarrow X$, let $\bar{\sigma}$ be the singular n -simplex obtained by preceding σ by the linear homeomorphism of $[v_0, \dots, v_n]$ reversing the order of the vertices. Thus $\bar{\sigma}(v_i) = \sigma(v_{n-i})$. This reversal of vertices is the product of $n + (n-1) + \dots + 1 = n(n+1)/2$ transpositions of adjacent vertices, each of which reverses orientation of the n -simplex since it is a reflection across an $(n-1)$ -dimensional hyperplane. So to take orientations into account we would expect that a sign $\varepsilon_n = (-1)^{n(n+1)/2}$ ought to be inserted. Hence we define a homomorphism $\rho: C_n(X) \rightarrow C_n(X)$ by $\rho(\sigma) = \varepsilon_n \bar{\sigma}$.

We will show that ρ is a chain map, chain homotopic to the identity, so it induces the identity on cohomology. From this the theorem quickly follows. Namely, the formulas

$$\begin{aligned} (\rho^* \varphi \smile \rho^* \psi)(\sigma) &= \varphi(\varepsilon_k \sigma| [v_k, \dots, v_0]) \psi(\varepsilon_\ell \sigma| [v_{k+\ell}, \dots, v_k]) \\ \rho^*(\psi \smile \varphi)(\sigma) &= \varepsilon_{k+\ell} \psi(\sigma| [v_{k+\ell}, \dots, v_k]) \varphi(\sigma| [v_k, \dots, v_0]) \end{aligned}$$

show that $\varepsilon_k \varepsilon_\ell (\rho^* \varphi \smile \rho^* \psi) = \varepsilon_{k+\ell} \rho^*(\psi \smile \varphi)$, since we assume R is commutative. A trivial calculation gives $\varepsilon_{k+\ell} = (-1)^{k\ell} \varepsilon_k \varepsilon_\ell$, hence $\rho^* \varphi \smile \rho^* \psi = (-1)^{k\ell} \rho^*(\psi \smile \varphi)$. Since ρ is chain homotopic to the identity, the ρ^* 's disappear when we pass to cohomology classes, and so we obtain the desired formula $\alpha \smile \beta = (-1)^{k\ell} \beta \smile \alpha$.

The chain map property $\partial\rho = \rho\partial$ can be verified by calculating, for a singular n -simplex σ ,

$$\begin{aligned} \partial\rho(\sigma) &= \varepsilon_n \sum_i (-1)^i \sigma| [v_n, \dots, \hat{v}_{n-i}, \dots, v_0] \\ \rho\partial(\sigma) &= \rho\left(\sum_i (-1)^i \sigma| [v_0, \dots, \hat{v}_i, \dots, v_n]\right) \\ &= \varepsilon_{n-1} \sum_i (-1)^{n-i} \sigma| [v_n, \dots, \hat{v}_{n-i}, \dots, v_0] \end{aligned}$$

which reduces us to the easily checked identity $\varepsilon_n = (-1)^n \varepsilon_{n-1}$.

To define a chain homotopy between ρ and the identity we are motivated by the construction of the prism operator P in the proof that homotopic maps induce the same homomorphism on homology, in Theorem 2.10. The main ingredient in the construction of P was a subdivision of $\Delta^n \times I$ into $(n+1)$ -simplices with vertices v_i in $\Delta^n \times \{0\}$ and w_i in $\Delta^n \times \{1\}$, the vertex w_i lying directly above v_i . Using the same subdivision, and letting $\pi: \Delta^n \times I \rightarrow \Delta^n$ be the projection, we now define $P: C_n(X) \rightarrow C_{n+1}(X)$ by

$$P(\sigma) = \sum_i (-1)^i \varepsilon_{n-i} (\sigma\pi)| [v_0, \dots, v_i, w_n, \dots, w_i]$$

Thus the w -vertices are written in reverse order, and there is a compensating sign ε_{n-i} . One can view this formula as arising from the Δ -complex structure on $\Delta^n \times I$ in which the vertices are ordered $v_0, \dots, v_n, w_n, \dots, w_0$ rather than the more natural ordering $v_0, \dots, v_n, w_0, \dots, w_n$.

To show $\partial P + P\partial = \rho - \mathbb{1}$ we first calculate ∂P , leaving out σ 's and $\sigma\pi$'s for notational simplicity:

$$\begin{aligned} \partial P = & \sum_{j \leq i} (-1)^i (-1)^j \varepsilon_{n-i} [v_0, \dots, \hat{v}_j, \dots, v_i, w_n, \dots, w_i] \\ & + \sum_{j \geq i} (-1)^i (-1)^{i+1+n-j} \varepsilon_{n-i} [v_0, \dots, v_i, w_n, \dots, \hat{w}_j, \dots, w_i] \end{aligned}$$

The $j = i$ terms in these two sums give

$$\begin{aligned} \varepsilon_n [w_n, \dots, w_0] + \sum_{i > 0} \varepsilon_{n-i} [v_0, \dots, v_{i-1}, w_n, \dots, w_i] \\ + \sum_{i < n} (-1)^{n+i+1} \varepsilon_{n-i} [v_0, \dots, v_i, w_n, \dots, w_{i+1}] - [v_0, \dots, v_n] \end{aligned}$$

In this expression the two summation terms cancel since replacing i by $i-1$ in the second sum produces a new sign $(-1)^{n+i} \varepsilon_{n-i+1} = -\varepsilon_{n-i}$. The remaining two terms $\varepsilon_n [w_n, \dots, w_0]$ and $-[v_0, \dots, v_n]$ represent $\rho(\sigma) - \sigma$. So in order to show that $\partial P + P\partial = \rho - \mathbb{1}$, it remains to check that in the formula for ∂P above, the terms with $j \neq i$ give $-P\partial$. Calculating $P\partial$ from the definitions, we have

$$\begin{aligned} P\partial = & \sum_{i < j} (-1)^i (-1)^j \varepsilon_{n-i-1} [v_0, \dots, v_i, w_n, \dots, \hat{w}_j, \dots, w_i] \\ & + \sum_{i > j} (-1)^{i-1} (-1)^j \varepsilon_{n-i} [v_0, \dots, \hat{v}_j, \dots, v_i, w_n, \dots, w_i] \end{aligned}$$

Since $\varepsilon_{n-i} = (-1)^{n-i} \varepsilon_{n-i-1}$, this finishes the verification that $\partial P + P\partial = \rho - \mathbb{1}$, and so the theorem is proved when $A = \emptyset$. The proof also applies when $A \neq \emptyset$ since the maps ρ and P take chains in A to chains in A , so the dual homomorphisms ρ^* and P^* act on relative cochains. $\square$

Example 3.15: Exterior Algebras. In general, the **exterior algebra** $\Lambda_R[\alpha_1, \alpha_2, \dots]$ over a commutative ring R with identity is the free R -module with basis the finite products $\alpha_{i_1} \cdots \alpha_{i_k}$, $i_1 < \dots < i_k$, with multiplication defined by the rules $\alpha_i \alpha_j = -\alpha_j \alpha_i$ and $\alpha_i^2 = 0$, the latter relation being a special case of the former if $2 \neq 0$ in R . The empty product of α_i 's is allowed, and provides an identity element 1 in $\Lambda_R[\alpha_1, \alpha_2, \dots]$. The calculation of cup products in the n -torus T^n in Example 3.11 shows that $H^*(T^n; R)$ is the exterior algebra $\Lambda_R[\alpha_1, \dots, \alpha_n]$. The same is true for any product of odd-dimensional spheres, as could be proved by a continuation of the calculation for the torus or by simply quoting Theorem 3.16 below.

A Künneth Formula

One might guess that there should be some connection between cup product and product spaces, and indeed this is the case, as we will show in this subsection.

To begin, we define the **cross product**, or **external cup product** as it is sometimes called. This is the map

$$H^*(X; R) \times H^*(Y; R) \xrightarrow{\times} H^*(X \times Y; R)$$

given by $a \times b = p_1^*(a) \smile p_2^*(b)$ where p_1 and p_2 are the projections of $X \times Y$ onto X and Y . Since cup product is distributive, the cross product is bilinear, that is, linear in each variable separately. We might hope that the cross product map would be an isomorphism in many cases, thereby giving a nice description of the cohomology rings of these product spaces. However, a bilinear map is rarely a homomorphism, so it could hardly be an isomorphism. Fortunately there is a nice algebraic solution to this problem, and that is to replace the direct product $H^*(X; R) \times H^*(Y; R)$ by the tensor product $H^*(X; R) \otimes_R H^*(Y; R)$.

Let us review the definition and basic properties of tensor product. For abelian groups A and B the tensor product $A \otimes B$ is defined to be the abelian group with generators $a \otimes b$ for $a \in A$, $b \in B$, and relations $(a + a') \otimes b = a \otimes b + a' \otimes b$ and $a \otimes (b + b') = a \otimes b + a \otimes b'$. So the zero element of $A \otimes B$ is $0 \otimes 0 = 0 \otimes b = a \otimes 0$, and $-(a \otimes b) = -a \otimes b = a \otimes (-b)$. Some readily verified elementary properties are:

- (1) $A \otimes B \approx B \otimes A$.
- (2) $(\bigoplus_i A_i) \otimes B \approx \bigoplus_i (A_i \otimes B)$.
- (3) $(A \otimes B) \otimes C \approx A \otimes (B \otimes C)$.
- (4) $\mathbb{Z} \otimes A \approx A$.
- (5) $\mathbb{Z}_n \otimes A \approx A/nA$.
- (6) A pair of homomorphisms $f: A \rightarrow A'$ and $g: B \rightarrow B'$ induces a homomorphism $f \otimes g: A \otimes B \rightarrow A' \otimes B'$ via $(f \otimes g)(a \otimes b) = f(a) \otimes g(b)$.
- (7) A bilinear map $\varphi: A \times B \rightarrow C$ induces a homomorphism $A \otimes B \rightarrow C$ sending $a \otimes b$ to $\varphi(a, b)$.

In (1)–(5) the isomorphisms are the obvious ones, for example $a \otimes b \mapsto b \otimes a$ in (1) and $n \otimes a \mapsto na$ in (4). Properties (1), (2), (4), and (5) allow the calculation of tensor products of finitely generated abelian groups.

The generalization to tensor product of modules over a commutative ring R is easy. One defines $A \otimes_R B$ for R -modules A and B to be the quotient of $A \otimes B$ obtained by imposing the further relations $ra \otimes b = a \otimes rb$ for $r \in R$, $a \in A$, and $b \in B$. This relation guarantees that $A \otimes_R B$ is again an R -module. In case R is not commutative, one assumes A is a right R -module and B is a left R -module, and the relation is written instead $ar \otimes b = a \otimes rb$, but now $A \otimes_R B$ is only an abelian group, not an R -module necessarily. It is an easy algebra exercise to see that $A \otimes_R B = A \otimes B$ when R